

MEX68USE

User System Evaluator (USE)

The MEX68USE User System Evaluator (USE), working with the M6800 EXORciser and the optional MEX68SA Systems Analyzer Module, provides the user with the complete systems development tool. That is, USE extends all of the EXORciser's EXbug functions and all of the optional Systems Analyzer functions into the user's system. The USE-equipped EXORciser provides the designer with the capability of configuring an emulation of his M6800 microprocessor system in the EXORciser, external to the EXORciser, or a combination of EXORciser-mounted modules working with the user's external system. The designer also can develop his system software and firmware on the EXORciser and can use the USE-equipped EXORciser to debug his system hardware and software. In fact, a USE-equipped EXORciser can even be used to test and evaluate the user's production systems.

- Extends the EXORciser EXbug functions into the user's system.
- Operates from the user's system clock or from the EXORciser's internal clock.
- Permits the user to share the EXORciser's memory and input/output modules with the user's system.
- Extends the optional Systems Analyzer Module's functions into the user's system when the Systems Analyzer is used with the EXORciser.
- Provides convenient block/fault isolation.
- Uses present commands — no new protocol to be learned.

USE consists of three assemblies: the USE Processor Module, the USE Intercept Module, and the USE Cable and Buffer Assembly. The *USE Processor Module* is used in place of the MEX6800 MPU Module in the EXORciser and provides the following functions:

- Shares its MC6800 MPU device between the EXORciser and the user's system.
- Provides the EXORciser internal clock circuitry.
- Provides the option of using the EXORciser clock within the EXORciser.
- Enables the overall system — user's system and EXORciser — to operate with the user's system clock.
- Interfaces the MC6800 MPU address, data, and control signals with the EXORciser bus.
- Interfaces the MC6800 MPU address, data, and control signals with the user's system via the USE Cable and Buffer Assembly.
- Interfaces selected MC6800 MPU signals (IRQ, NMI, HALT, VMA, and BA) with the optional Systems Analyzer Module via the USE Intercept Module.

The *USE Intercept Module* plugs directly into the EXORciser bus and is used to mount the optional Systems Analyzer Module. A cable assembly couples the selected control signals between the Systems Analyzer and the USE Processor Module.

The *USE Cable and Buffer Assembly* connects the USE Processor Module into the user's system. A 40-pin connector on this assembly plugs into the MPU socket in the user's system permitting the USE Processor Module's MPU to control the operation of the user's system. In this application everything within the EXORciser appears to be within the MPU in the user's system. Through special jumper connections, this assembly also can transfer the timing signals required by the dynamic memories in the EXORciser to operate with the user's system clock. In addition, the USE Cable and Buffer Assembly buffers the transferred signals.

MEX68USE

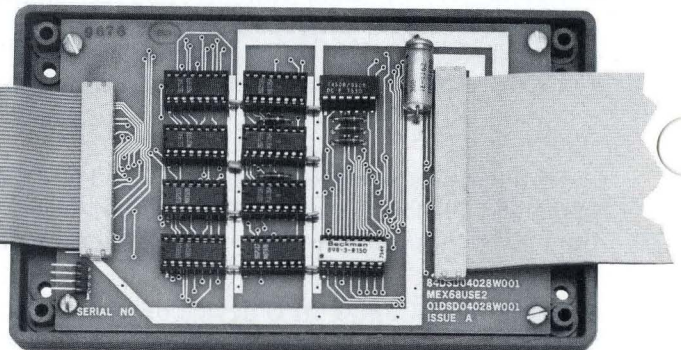
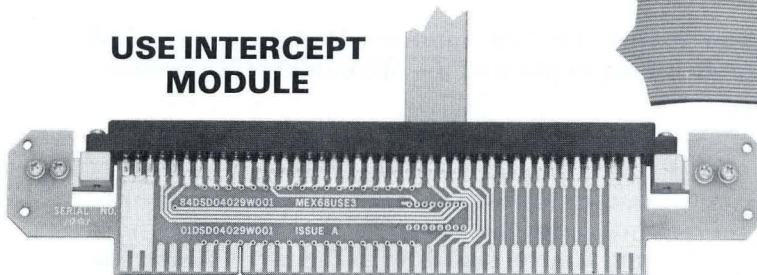
User System Evaluator (USE)

Specifications

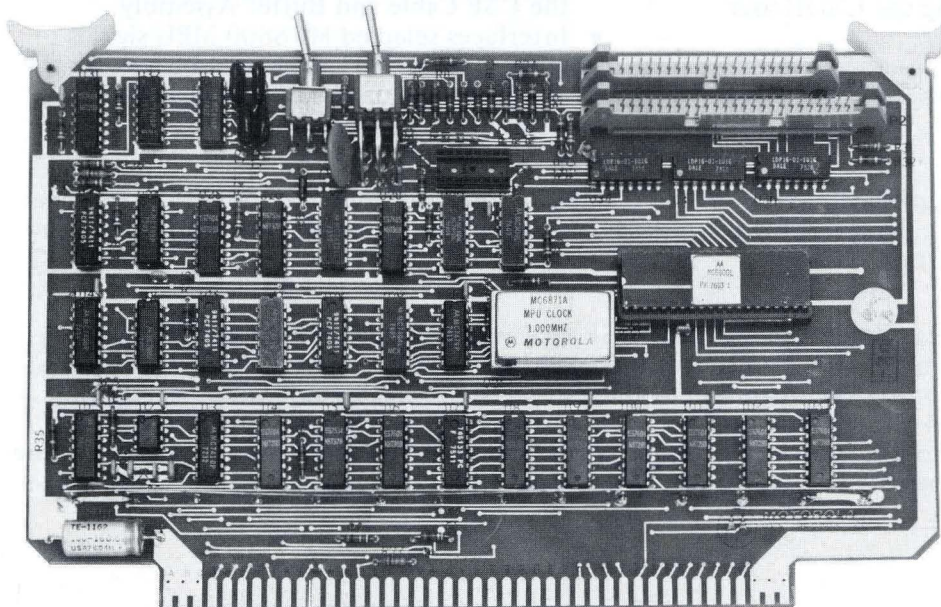
Microprocessor	MC6800 MPU
Word Size	
Data	8 bits
Address	16 bits
Instruction	8, 16, or 24 bits
Instruction Set	72 variable length instructions
Addressing Modes	Seven address modes – Direct, Relative, Immediate, Indexed, Extended, Implied, and Accumulator
Memory Size Capability	65,536 bytes with the EXORciser EXbug Firmware (when used) using 4K bytes. Addresses EXORciser ROM, RAM, and I/O and user's system ROM, RAM, and I/O in any combination.
Clock Signals	Switch Selectable Crystal controlled 1 MHz clock with capability to work with dynamic and slow memories. Operates in the EXORciser system. or User's system clock operating in the EXORciser and the User's system.

Operating Temperatures	0 to 70°C.
Power Requirements	+5 Vdc at 1.5 A
Physical Characteristics	
USE Processor Module	
Width	9.75 in.
Height	5.75 in. above connector
Board Thickness	0.062 in.
USE Intercept Module	
Width	9.75 in.
Height	1.875 in.
Board Thickness	0.062 in.
Cable Assembly	
Length	1 ft.
USE Cable and Buffer Assembly	
Case Height	0.875 in.
Case Length	7.50 in.
Case Width	4.32 in.
Cable Assembly	
Length	2 cables 5 ft. each 1 cable 13 in.

**USE INTERCEPT
MODULE**



**USE CABLE AND
BUFFER ASSEMBLY
(Cover Removed)**



**USE
PROCESSOR
MODULE**



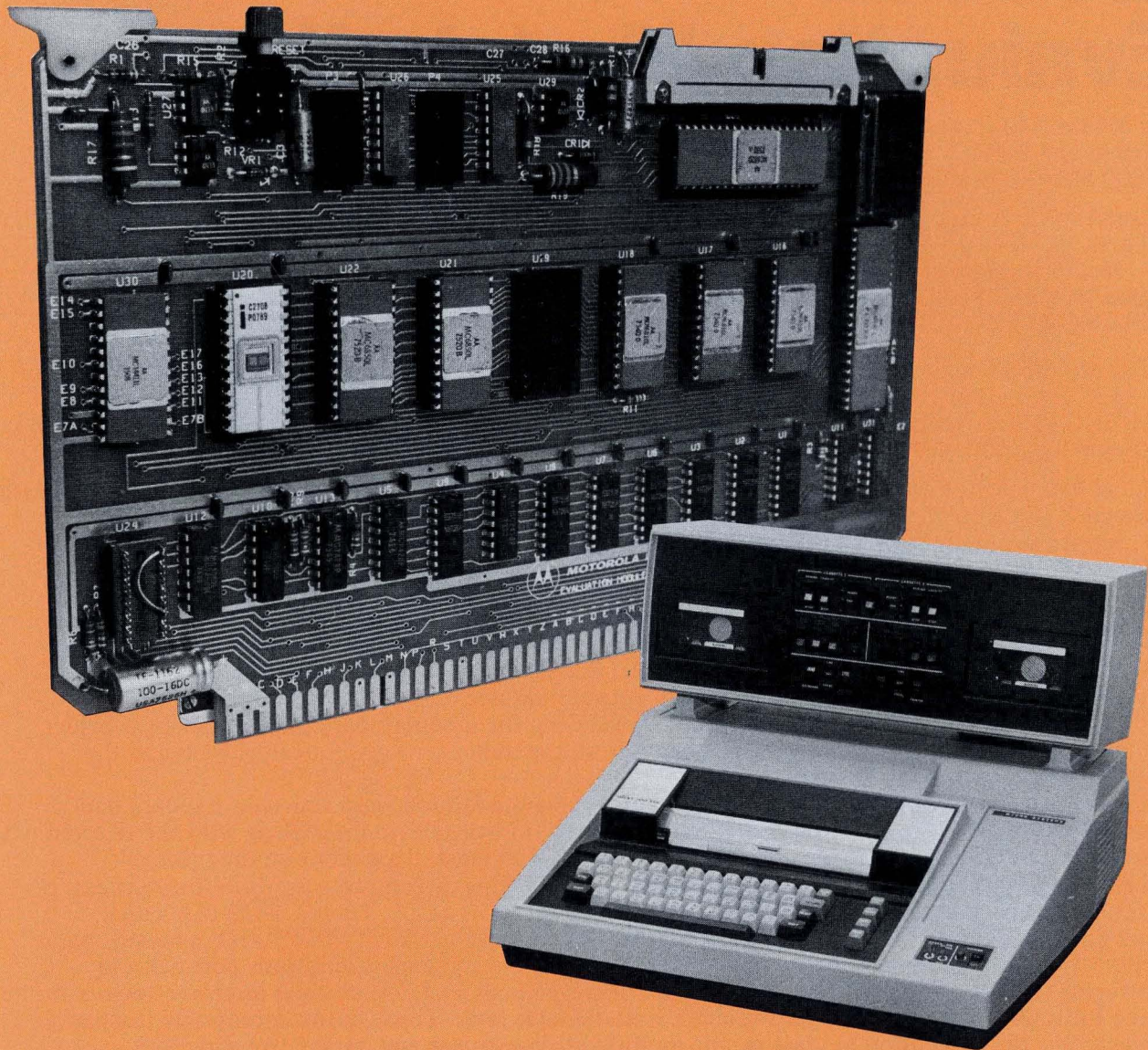
MOTOROLA Semiconductor Products Inc.

BOX 20912 • PHOENIX, ARIZONA 85036 • A SUBSIDIARY OF MOTOROLA INC.

10211 PRINTED IN USA 7-76 IMPERIAL LITHO B58554



MOTOROLA



M6800B Evaluation Module II

For Hardware Evaluation and Program Development

M6800B

Evaluation Module II

The M6800B Evaluation Module II is a Microcomputer that provides the user with an efficient and economical off-the-shelf means to evaluate the operating characteristics of the M6800 Microcomputer family of parts in various applications. It contains the following components:

- One M6800 Microprocessing Unit (MPU)
- Three MCM6810 128 x 8 Random Access Memories (RAM)
- One MCM6820 Peripheral Interface Adapter (PIA)
- One MCM6830 1024 x 8 Read Only Memory (ROM)
- Two MC6850 Asynchronous Communications Interface Adapters (ACIA)
- One MC14411 Bit Rate Generator (BRG)
- Data and address buffers

The M6800B Evaluation Module II interfaces with either a TTY (20 mA neutral current loop) or an RS-232C compatible terminal. This data terminal provides direct communication with the module's MINIBUG control program. The MINIBUG control program, stored in the module's read only memory, in conjunction with the selected data terminal, provides the following functions:

- Load data into the Evaluation Module
- Display and, if required, change the data in the module's random access memory
- Display and, if required, change the contents of the MPU registers
- Print out or punch (record) on tape the data stored in the module's memory
- Set the number of stop bits for data communication
- Run the user's program

The Evaluation Module provides the user with 256 bytes of random access memory on which to develop his programs. In addition, this module, through its address and data buffers, is capable of interfacing with additional memory and input/output modules.

The module's clock circuitry, in addition to generating the module's basic timing signals, provides the Evaluation Module with dynamic memory refresh capability on a cycle stealing basis and the ability to work with slow memories.

The MC14411 Bit Rate Generator provides the Evaluation Module with fourteen standard baud rates — six of which are jumper selectable. The MINIBUG II Firmware, in interfacing with the selected data terminal, is capable of transferring data at either 110 or 300 baud, dependent on the terminal.

The Evaluation Module also has the capability of connecting directly to a user's defined process or peripheral device through its MC6820 PIA and second MC6850 ACIA. The MC6820 PIA's dual input/output ports and control lines permit interfacing the Evaluation Module with keyboards, basic printers, displays, and similar devices. The MC6850 ACIA's serial communications port permits interfacing the Evaluation Module with modems, data terminals, and similar serial interface devices at one of the Bit Rate Generator's jumper selectable data transfer rates. Through the appropriate jumper connections, the Evaluation Module can be modified to operate at any one of 14 baud rates.

Options

Option	Part Number	Description
1	MEC6800B	Assembled and Tested Evaluation Module Assembly
2	MEC6800B1	Bareboard — Printed Circuit Board only. User supplies components.

Parts supplied with the MEC6800B — Three 16-pin component boards.
PIA Connector and Flatribbon Cable.
M6900B Evaluation Module II User's Guide.

M6800B Evaluation Module II

Specifications

Power Requirements	+5 Vdc @ 2 A +12 Vdc @ 50 mA -12 Vdc @ 50 mA 921.6 kHz
Clock Frequency	
Signal Characteristics	
Connector P1	
Address Bus	Three-State TTL Voltage Compatible
Logic "0"	0.5 V max at 40 mA through a resistor to V _{CC}
Logic "1"	2.6 V min at -10 mA through a resistor to ground
Output Off-State Leakage Current	100 μ A max at 2.6 V
Data Bus	TTL Voltage Compatible
Input Logic "0"	0.0-0.85 V (-200 μ A max at 0.4 V)
Input Logic "1"	2.0-5.25 V (25 μ A max at 5.25 V)
Output Logic "0"	0.5 V max at 40 mA through a resistor to V _{CC}
Output Logic "1"	2.6 V min at -10 mA through a resistor to ground
Control Bus	
Input Logic "0"	TTL Voltage Compatible
Input Logic "1"	TTL Voltage Compatible
Output Logic "0"	0.5 V max at 40 mA through a resistor to V _{CC}
Output Logic "1"	2.6 V min at -10 mA through a resistor to ground
Connector P2, PIA Signals	
Data Signals	
PA0-PA7 Input/Output Lines	TTL Voltage Compatible
PB0-PB7 Input/Output Lines	Three-State TTL Voltage Compatible
Control Lines	
CA1, CA2, and CB1	TTL Voltage Compatible
CB2	Three-State TTL Voltage Compatible
Connector P3, Terminal Interface Signals	
Data Transfer Rate	110 or 300 Baud
Data Signal Characteristics	TTY (20 mA neutral current loop) or EIA RS-232C compatible
Reader Control Signal	Control signal for manual TTY devices modified for external control
Connector P4, Terminal Interface Signals	
Data Transfer Rate	110, 134.5, 300, 600, 1200 and 2400 Baud (with jumper modifications 75, 150, 200, 1800, 3600, 4800, 7200 and 9600).
Data Signals	EIA RS-232C compatible

Connections and Descriptions

Connector P1 — Data, Address, and Control Bus

Signal Mnemonic	Signal Name and Description
IRQ	Interrupt Request — This input signal requests that an interrupt sequence be generated within the MPU. The processor will wait until it completes the instruction that is being executed before it recognizes the request. Then, if the interrupt mask bit in the condition code register is not set (interrupt not masked), the MPU will begin an interrupt sequence. The MPU will respond to the interrupt request by setting the interrupt mask bit high so that no further interrupts may occur.
NMI	Non-Maskable Interrupt — This input signal requests that a non-maskable interrupt sequence be generated within the processor. The processor will complete the instruction that is being executed before it recognizes the NMI signal.
VMA	Valid Memory Address — This output indicates to the memory and peripheral devices that the address on the address bus is valid.
ϕ 2	Phase 2 Clock signal — This signal is derived from the memory clock and used to synchronize the transfer of data on the data bus.
MEMCLK	Memory Clock — This is the basic 921.6 kHz clock signal used by the Evaluation Module to generate its ϕ 1 and ϕ 2 clock signals. A dynamic memory device would use this signal to control its timing.

Connections and Descriptions

Connector P1 — Data, Address, and Control Bus (continued)

Signal Mnemonic	Signal Name and Description
TSC	Three-State Control — This input, when high, causes all of the address lines and read/write line to go to their off, or high-impedance, state. The valid memory address, valid user's address, and the bus available signals will be forced low. The data bus is not affected by the three-state control.
BA	Bus Available — This output signal will normally be a low level. When activated, it will go high indicating that the address bus is available. This will occur if the <u>Ha</u> lt line is low or the MPU is in the WAIT state as the result of executing a WAIT instruction.
MEMRDY	Memory Ready — This signal enables the Evaluation Module to work with slow memories. The Evaluation Module, on receiving a low level Memory Ready input, stops generating the $\phi 1$ and $\phi 2$ clock signals with $\phi 2$ high.
$\overline{D0-D7}$	Data Bus — These bi-directional lines, when enabled, transfer data between the MPU and any external modules.
A0-A15	Address Bus — This address line, when enabled transfers the MPU program counter output to any external modules.
<u>HALT</u>	<u>Ha</u> lt — When this input is in the high state, the MPU will fetch the instruction addressed by the program counter and start execution. When low all activity in the MPU will be halted.
<u>RESET</u>	<u>Re</u> set — This line goes low when the RESET switch is actuated and may be used to reset any external devices. The RESET switch, on being actuated, initiates a MC6800 MPU cold-start vectored interrupt routine.
R/W	Read/Write — This MPU output signal indicates to the external modules whether the MPU is in a read (high) or a write (low) state. The normal standby state of this signal is read (high).
<u>REFREQ</u>	<u>Re</u> fresh <u>Re</u> quest — This signal, when present, initiates a memory refresh operation. The Evaluation Module, on receiving this input from an external module, stops generating the $\phi 1$ and $\phi 2$ clock signals with $\phi 1$ high and, through the Refresh Grant command, instructs the dynamic memory modules to refresh their memories.
REFGNT	<u>Re</u> fresh <u>Gr</u> ant — This signal instructs the dynamic memory modules to refresh their memories.
VUA	Valid User's Address — This signal indicates to the memory and peripheral devices that the address on the bus is valid. On Evaluation Module II, VUA is the same signal as VMA.

Connector P2 — Peripheral Interface Adapter Signals

Evaluation Module II provides the user with the capability of connecting a parallel peripheral device to the module's MC6820 PIA. This PIA has dual 8-bit input/output ports — PA0-PA7 and PB0-PB7 — along with four control lines — CA1, CA2, CB1, and CB2. Refer to the M6800 System Design Data for details on these signals.

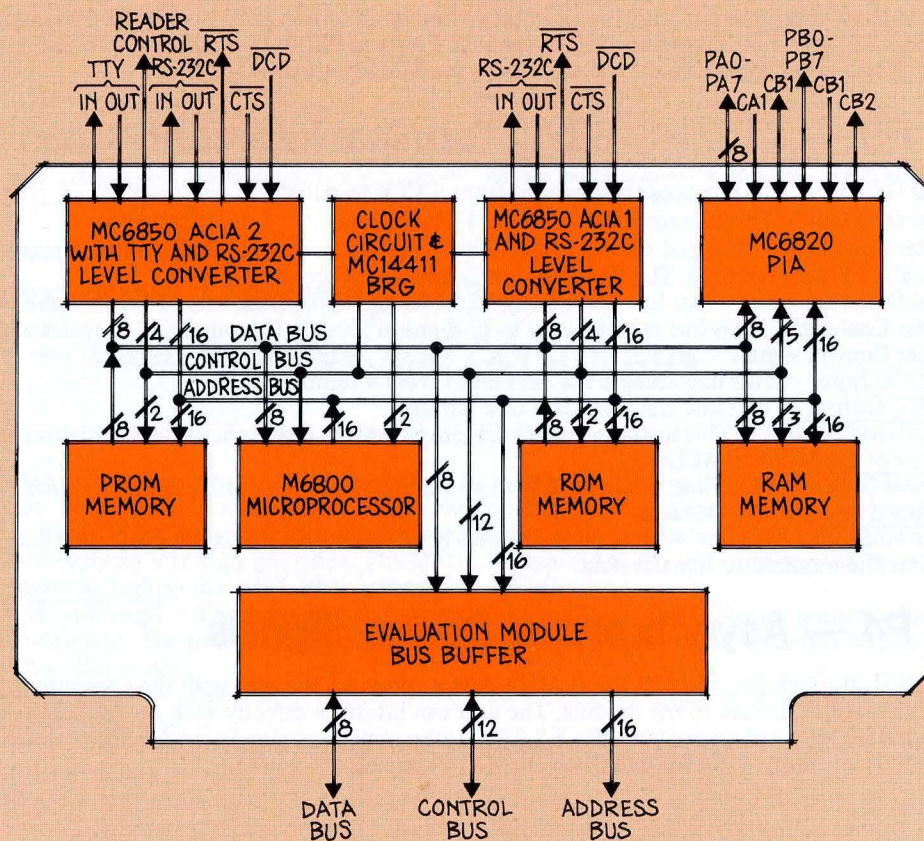
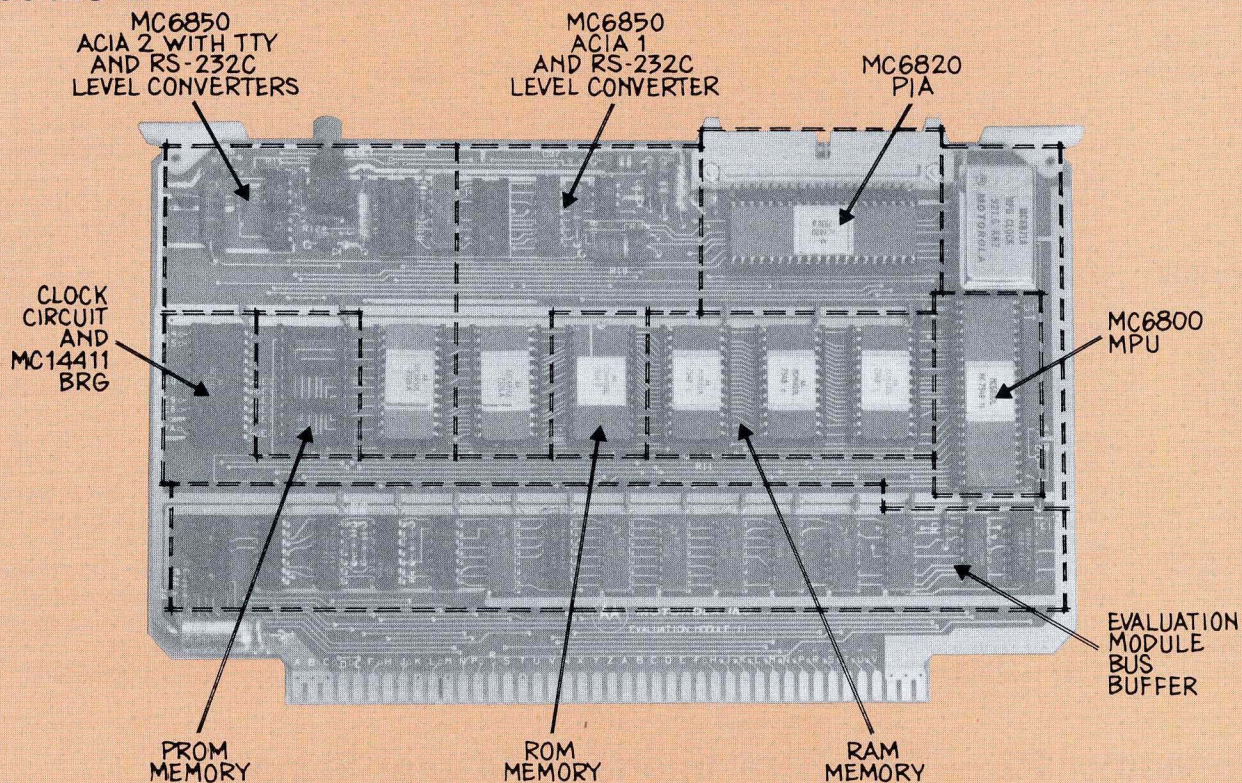
Connector P3 — TTY/RS-232C Terminal Interconnection Signals

TTY IN	Serial Data In — This line accepts the input from a TTY terminal.
TTY OUT	Serial Data Out — This line transfers data to a TTY terminal.
—	Reader Control — This signal provides the control required to operate the paper tape reader on a modified manual TTY data terminal. The terminal now is under MPU control.
—	Serial Data Common — This line provides a return for the Serial Data In and Serial Data Out signals.
—	Reader Common — This line is connected to Evaluation Module II ground and provides a return for the Reader Control signal.
RS-232 IN	RS-232C Input — This line accepts the data input from a terminal.
RS-232 OUT	RS-232C Output — This line transfers data to a terminal.
DCD	Data Carrier Detect — This high impedance TTL input, when high, inhibits and initializes the receiver section of the MC6850 ACIA.
<u>RTS</u>	<u>Re</u> quest to <u>Se</u> nd — This line, when used with an RS-232C terminal, informs the terminal that Evaluation Module is ready to receive data.
<u>CTS</u>	<u>Cl</u> ear to <u>Se</u> nd — This line is a low level when an RS-232C terminal is connected to Evaluation Module II and informs the terminal to transfer data.

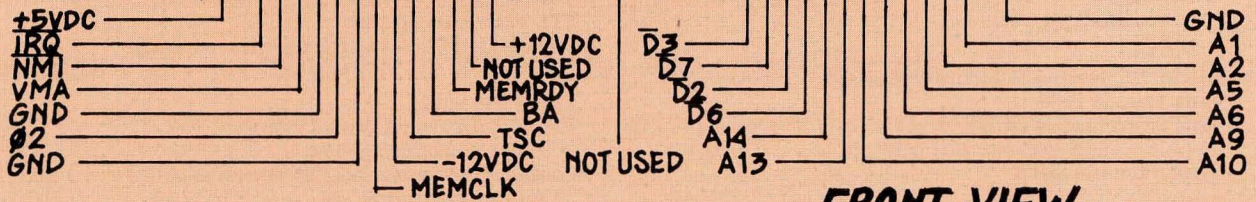
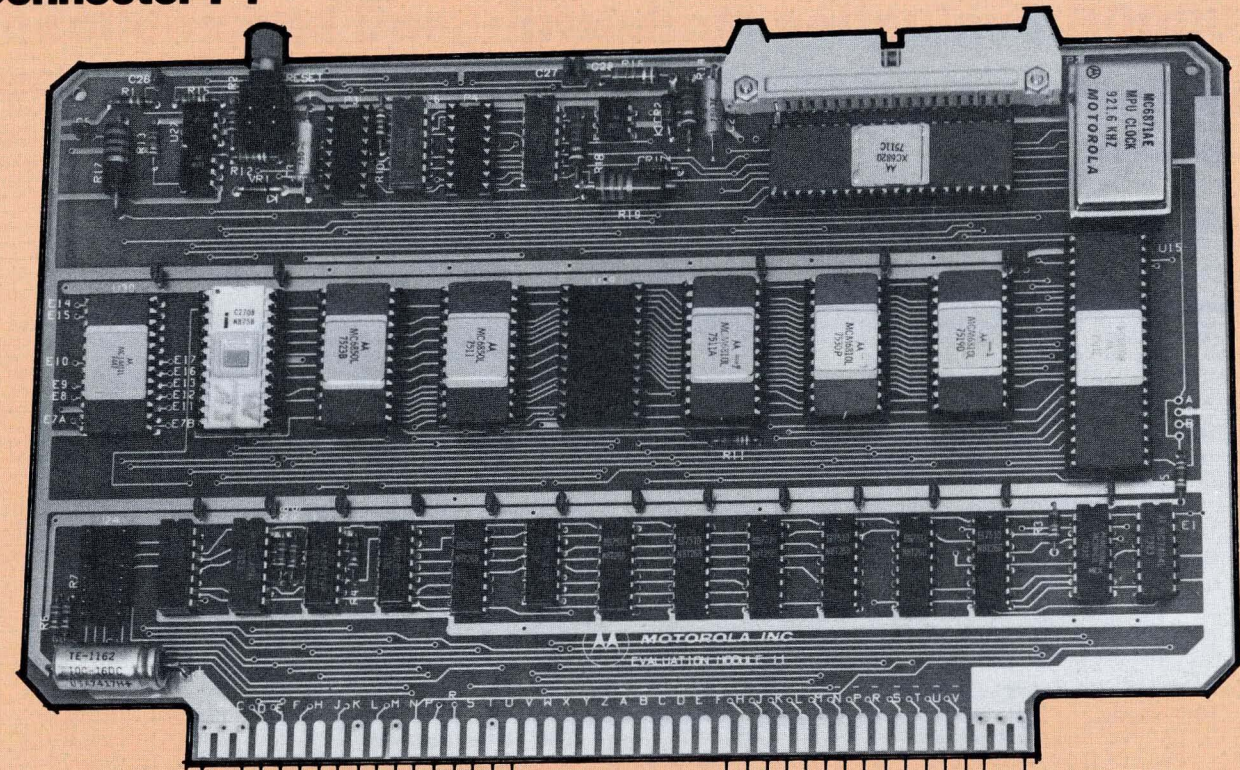
Connector P4 — Asynchronous Device Signals

Evaluation Module II, through the other MC6850 ACIA device, provides the user with the capability of connecting asynchronous communications devices to the module. The user can interface directly with the module's ACIA or through the module's RS-232C level converters. The RS-232C interconnection signals are described above. Refer to the M6800 System Design Data for details on the ACIA's signals.

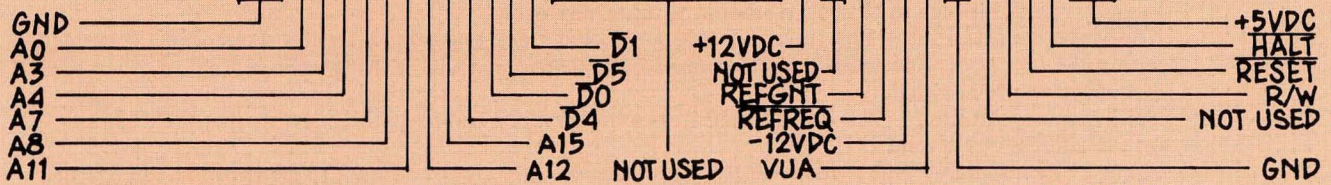
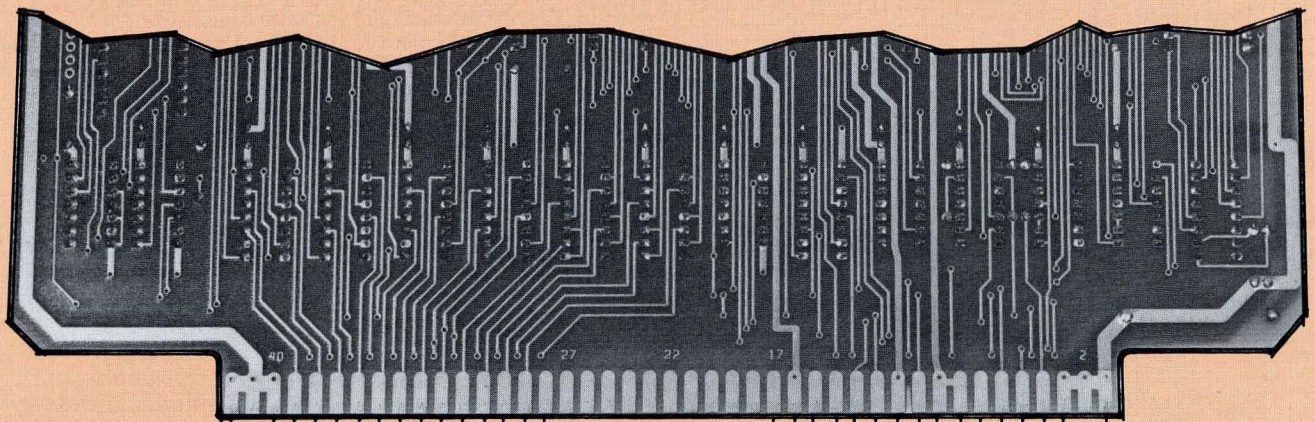
Evaluation Module II



Connector P1



FRONT VIEW



REAR VIEW

**MOTOROLA Semiconductor Products Inc.**

BOX 20912 • PHOENIX, ARIZONA 85036 • A SUBSIDIARY OF MOTOROLA INC.

10211 PRINTED IN USA 5-76 IMPERIAL LITHO B57297 20M



MOTOROLA MICROSYSTEMS

The only peripheral required to use the EXORciser for design purposes is a data terminal. For major design efforts, however, considerable design time can be saved with additional peripherals that speed up program development. EXORDisk and EXORTape are among these development tools.

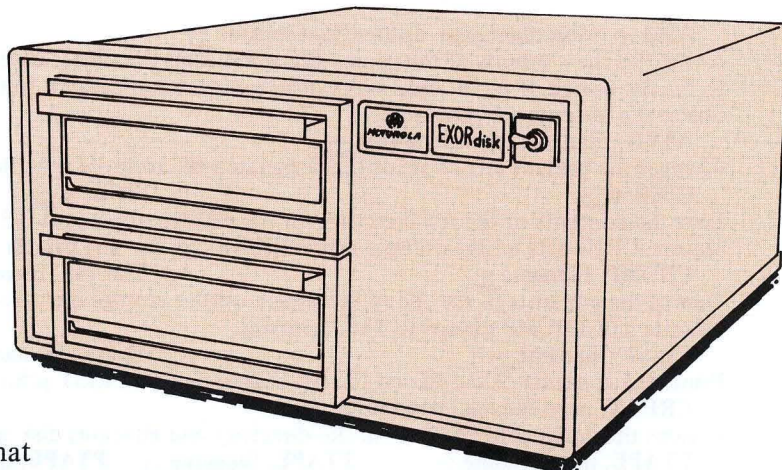


M6800 Support Peripherals



MOTOROLA MICROSYSTEMS

M6800 EXORdisk



The EXORdisk is a "floppy disk" storage system that extends the EXORciser capacity by up to one-million bytes of memory. Together with the EXORciser, and a separately available Interface Module, it provides a complete program-development system whose high-speed capabilities can be matched only by much more expensive minicomputers. The floppy disk system allows the user to assemble, in 30 seconds, programs that would require up to 2½ hour assembly time with a standard ASR-33 teletype reader.

The EXORdisk system consists of the disk-drive mechanism and a diskette which houses the EXORciser Flexible Disk Operating System II, EDOS II. The latter controls the EXORdisk and interface operations. Functions include the loading of data from the EXORdisk to the EXORciser memory or different diskettes and between diskettes and other storage media such as paper tape and cassette. If desired, the EDOS II program may be purchased on a separate diskette which contains, in addition, the M6800 Resident Editor and Assembler programs. This permits disk-to-disk program editing and assembling in addition to the various transfer functions.

The EXORdisk connects to the EXORciser via an interface module that is an optional component of the EXORciser. This module can be used to simultaneously connect an EXORTape peripheral to the EXORciser.

The EDOS II program will handle and control up to four disk drives, thereby offering memory capacity ranging from 256,256 eight-bit bytes to over 1-million bytes.

Specifications

Diskette Format:

2,050,048 Bits/Diskette
256,256 Bytes/Diskette
77 Tracks/Diskette
26 Sectors/Track
128 Bytes/Sector

Performance:

Rotational Speed – 360 RPM
Track-To-Track Access Time – 10 ms
Head Load – 40 ms
Sector Read/Write Time – 5 ms
Average Latency Time – 83 ms
Interrecord Time – 1 ms
Head Unload Time Out – 700 ms
(for maximum head and media life)

Power Requirements: 115/220 Vac, 50/60 Hz, 200 Watts

Physical Characteristics: (2-drive system)

L x H x D 17 x 8 3/4 x 17 inches
Weight 65 pounds

Electrical:

All signals are TTL, Low-True
16 Output Lines, 8 Input Lines

Options and Ordering Information

Part Number	Description
M68FD3602	EXORdisk with two drives (includes appropriate two-drive cable)
M68FD3602-12	EXORdisk for 220 V, 50/60 Hz operation
M68FD3602-19	19-inch rack mounting kit
M68FD3602-24	Cable for expanding from two-drive to four-drive system.
M68IFC	M6800 EXORciser Interface Module
M68XAE6812D	Diskette containing EDOS II, M6800 Resident Editor and M6800 Resident Assembler

EXORdisk Commands

ASM, p, sourcefilename, destinationfilename ↷

Assembles the contents of the source file and directs the object to the destination file. If p = 3, only a listing is generated to the list device. If p = 4, only a hex object output is generated to the output object file. If p = 2, both listing and object file outputs are generated.

ATTR, filename, newattributes ↷

Changes the present attributes of the designated file to those specified in the new attributes field.

CDIR:u ↷

PDIR: u ↷

Lists the contents of the file directory on the diskette drive unit 0-3. Lists the filenames, attributes, and file size in sectors. CDIR lists on the console and PDIR lists on the line printer.

CDUMP, filename ↷

TDUMP, filename ↷

Dumps the contents of the file to the punch output storage device or the communication link device. CDUMP dumps to cassette and TDUMP dumps to TTY terminal.

CLIST, filename ↷

PLIST, filename ↷

Prints the contents of the file on the list output device. CLIST prints on the console and PLIST prints on the line printer.

CREAT, newfilename, newfilesize ↷

Creates the designated filename in the directory and allocates disk space equal to the size.

CTAPE, newfilename ↷

TTAPE, filename ↷

PTAPE, filename ↷

Loads the contents of the reader device into the specified file name on diskette. CTAPE transfers from cassette, PTAPE transfers from a high speed paper tape reader, and TTAPE transfers from a TTY terminal.

CXGEN ↷

PXGEN ↷

TXGEN ↷

Enables system generation of other EDOS versions that may become available in the future.

DUP ↷

Copies the contents of the diskette in drive unit 0 onto the diskette in drive unit 1.

EDIT, inputfilename, newoutputfilename ↷

EDIT, newoutputfilename ↷

Enables editing of the input file's contents. Edited data is stored into the output file.

EXBUG ↷

Returns to the EXORciser EXbug control program.

HOME: u ↷

Positions the disk head on drive unit u (0-3) to track 0.

INIT: u ↷

INITX ↷

Initiates the file directory on the diskette in drive unit u. Clears any existing user files on the diskette.

INITX implies drive unit 0 and u in INIT: u designates drive unit 1, 2, or 3.

LOAD, objectfilename ↷

Loads the contents of the file into RAM memory for execution.

MERGE, newfilename, filename1, filename2... filenamen ↷

Creates a new file which is a concentration of filenames 1-n, in that order.

PURGE: u, filename1, filename2... filenamen ↷

Deletes the designated files from the diskette in drive unit u (0-3) and then repacks the contents of that diskette, making disk space available for additional files.

RENAM, oldfilename, newfilename ↷

Renames the old file with the new file name.

Interface Command and Data Words

all signals are compatible with MC6820 PIA Chips

Input Data and Status: 8 bits, negative true, PIA address
ECO0, bits 0-7
Logic 1: 0 to 0.4 volts
Logic 0: 2.4 volts min

Output Data: 8 bits, negative true, PIA address
ECO6, bits 0-7.

Output Commands: 8 bits, negative true, PIA address
ECO2, bits 0-7.

Note: IRQA1 bit 7 is device BUSY.

COMMAND FUNCTIONS

SEEK AND VERIFY: Seeks Selected Track and Verifies Track
Address from ID Field:

SEEK TRACK 0: Seeks Track 0

SECTOR AND UNIT SELECT: Specifies Sector and Unit
Number for Read/Write Operation

TRACK SELECT: Specifies Selected Track to be used by
Next Seek

WRITE: Write Contents of Write Buffer to Selected Sector
and Unit on existing track

READ: Reads Contents of Selected Sector into Read Buffer

WRITE DELETED DATA ADDRESS MARK: Same as Write
but uses header in Data Field which can later be detected in
Read Operation

READ CRC: Same as Read but no data is transferred to the
Read Buffer. Used to verify integrity of data previously written

SHIFT WRITE BUFFER: Loads Data into Write Buffer

SHIFT READ BUFFER: Inputs Data from Read Buffer

GATE STATUS: Gates Status or Data onto Input Data Lines.

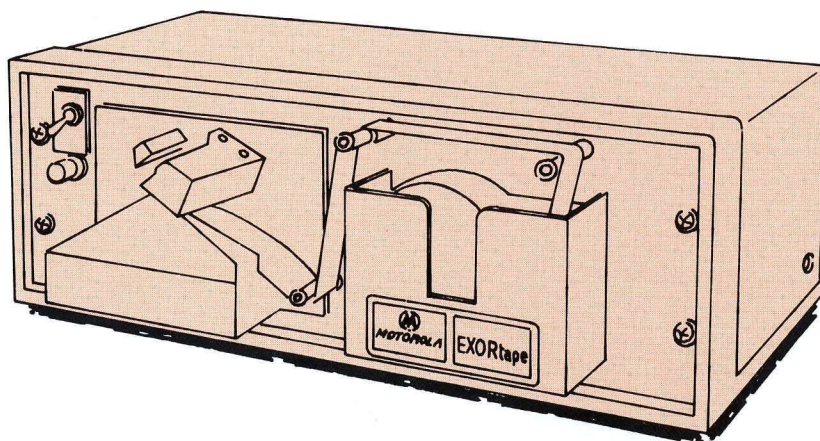


MOTOROLA Semiconductor Products Inc.

BOX 20912 • PHOENIX, ARIZONA 85036 • A SUBSIDIARY OF MOTOROLA INC.

M6800 EXORtape

- Plug Compatible with the EXORciser
- Compatible with the EXORciser's EXbug Firmware and Resident Software
- Up to 25 Times Faster Than a Standard ASR-33 Teletype Reader
- Photo Electric Character Detector for More Reliable Loading
- Derated Long Life, Line Filament Type Lamp



EXORtape is a high-speed paper-tape reader that provides an efficient means for program loading, editing and assembling. It is capable of loading up to 250 characters per second into EXORciser memory. The EXORtape connects to the EXORciser through the optional M6800 EXORciser Interface Module and is completely compatible with the EXORciser EXbug firmware and the M6800 Resident Software. Its Interface program also permits the user to daisy-chain other peripherals, such as the EXORDisk, into the system.

Specifications

Electrical	
Input Data	8 bits
Status Input	Reader busy
Start Command	HPST start bit
Power Requirements	115 Vac, 60 Hz single phase, 40 W
Optional	220 Vac, 50/60 Hz single phase
Manual Advance	Front panel push button
Speed	Up to 250 characters per second
Physical	
Cables	One 40 line ribbon cable, length 5 feet
Dimensions	
L x H x D	17-1/2 x 6 x 8-1/2"
Mounting	Stand alone, rubber feet, optimal 19 inch RETMA rack mount
Weight	10 pounds
I/O Cabling	Rear/Center
Tape Holder	Up to 5 inches diameter tapes

Options and Ordering Information

Part Number	Description
M68R680	Table Top EXORtape
M68R680-12	EXORtape with 220, 50/60 Hz Power Requirement
M68R680-19	19-inch Rack Mounting EXORtape
M68IFC	M6800 EXORciser Interface Module